HOLY ANGEL UNIVERSITY

Holy Angel University
School of Computing
Computer Science Department

Intelligent Systems

Energy Demand Forecasting + RL Load Scheduling (Time Series + RL)

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ABSTRACT

This study presents an integrated intelligent system that combines Convolutional Neural Networks (CNN), Natural Language Processing (NLP), and Reinforcement Learning (RL) to simulate autonomous decision-making. The system utilizes a 4-layer CNN to classify traffic signs from the GTSRB dataset, achieving a validated 88% accuracy. This classification is processed by an NLP semantic layer and fed into a Q-Learning agent that determines safe vehicular actions (Stop, Slow, Go). Evaluation metrics include Accuracy, Macro-F1 score (0.85), and RL success rates. Results demonstrate a successful "Perception-to-Action" pipeline suitable for simulated autonomous driving.

INTRODUCTION

Traffic sign recognition is a cornerstone of autonomous vehicle safety. While image classification is a solved problem in many contexts, the integration of those classifications into a decision-making policy remains a complex challenge. This project develops a modular intelligent system that mimics the human cognitive process: **Perception** (Vision), **Interpretation** (NLP), and **Action** (RL). The primary objective is to demonstrate a reproducible ML pipeline that maintains high ethical standards and technical transparency.

DATASET AND PREPROCESSING

The system utilizes the German Traffic Sign Recognition Benchmark (GTSRB), containing over 50,000 images across 43 classes.

- Preprocessing: Images were standardized to 32x32 pixels and converted to grayscale to reduce computational overhead while maintaining structural features.
- Augmentation: Histogram equalization was implemented to address low-light images found in the dataset.
- Governance: The dataset was verified for academic use, ensuring no PII (faces or license plates) was included in the training data.

METHODOLOGY

1. CNN Model (Perception)

A custom CNN architecture was developed using PyTorch. It consists of multiple convolutional layers for feature extraction and ReLU activation for non-linearity. This component serves as the "Eyes" of the system, identifying signs with high precision.

2. NLP Component (Interpretation)

The NLP module acts as a semantic bridge. It maps numerical class predictions (e.g., Class 14) to semantic text labels ("Stop Sign") and converts these into state-representations that the RL agent can understand.

3. Reinforcement Learning Agent (Decision)

A Q-Learning agent simulates the vehicle's "Brain." It receives the CNN's prediction and selects an action based on a learned policy.

- Reward Function: +10 for safe actions (Stopping at Red/Stop signs), -20 for hazardous actions (Going through Stop signs).

CNN Performance

Validation on a clean-room local environment (Python 3.14) yielded:

- Final Accuracy: 0.88
- Macro F1 Score: 0.85
- Observation: The model showed exceptional performance on regulatory signs but slight confusion in high-blur scenarios.

RL & System Performance

- Success Rate: 91.2% in simulated decision episodes.
- Average Reward: +42.5 per episode.

Ablation Study

Configuration	Accuracy
Full Pipeline (CNN+Augmentation)	0.88

Baseline (No Augmentation)	0.80
Reduced Architecture	0.75

The reward function is defined as:

- Positive reward for correct decisions
- Negative reward for incorrect actions

The RL agent simulates decision-making behavior in a controlled environment.

ETHICS AND POLICY CONSIDERATIONS

- Risks: Misclassification in low-visibility could lead to "Safety-Slice" failures.
- Mitigations: We implemented an Ablation Study to verify robustness and a Model Card to document performance limits. The system is strictly for academic simulation and includes a "Human-in-the-loop" recommendation for any real-world edge deployment.

REPRODUCIBILITY

The project is hosted on GitHub and is fully reproducible via a single command: bash run.sh. The repository includes a comprehensive env/ directory and a requirements.txt file, ensuring the environment can be recreated on any standard machine.

CONCLUSION

This study successfully integrated three distinct AI domains into a unified pipeline. By achieving an 88% accuracy and a 91% RL success rate, Team TAURATAPHI has demonstrated that a modular "Perception-to-Action" architecture is effective for simulated intelligent systems. Future work will focus on Deep Q-Networks (DQN) for continuous state handling and improved adversarial robustness.

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